# Speaker Notes

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**Slides**: 53 · **Duration**: 60 min · **Venue**: ARTIN

# 1 1. Building AI Systems

**On screen:**

> *ARTIN · 2026-08-18*
>
> *Reliability is engineered, not prompted.*
>
> *Ondrej (Ondra) Krajicek me@ondrejkrajicek.com · linkedin.com/in/OndrejKrajicek Built 2026-08-16 14:07 UTC*

Thank you for coming — none of you had to be here, which is the best possible audience to argue in front of. [speculative] That narrow gap — the architecture around the model, not the model — is the whole talk. [speculative] Every load-bearing claim carries a mechanism, a number or a named source — because this room does not buy adjectives. [speculative]

# 2 2. Reliability is engineered, not prompted

**On screen:**

> *The claim*
>
> *Building reliable AI systems is an engineering discipline, not a prompting one. [speculative]*
>
> - *Reliability comes from **external grounding** and **bounded autonomy** — not from asking the model to check its own work. [speculative]*
> - *The two load-bearing words are external and bounded: a check the model cannot grant itself, and autonomy inside a fence someone else drew. [speculative]*

Here is the thesis on slide two, so you know exactly what you came to agree or disagree with. [speculative] Building reliable AI systems is an engineering discipline, not a prompting one — a testable claim, not a slogan. [speculative] The hour hangs on two words: *external*, a check the model cannot grant itself, and *bounded*, autonomy inside a fence someone else drew and the model cannot move. [speculative] So push on it: if by the end you think these are not the load-bearing variables, tell me which ones are. [speculative]

# 3 3. Roadmap

**On screen:**

> *Roadmap*
>
> 1. *Prompt engineering stopped being the interesting problem*
> 2. *The five pillars of a reliable AI system*
> 3. *Loops — why self-verification underperforms*
> 4. *The harness — the system around the model*
> 5. *What this means for your Monday*

Five sections, roughly ten minutes each; no claim on this slide, it is the map. [speculative] First, why prompt engineering — the thing everyone got good at — stopped being the interesting problem. [speculative] Then the five pillars, where I spend the most time on grounding, because that is where the strongest number in the talk lives. [speculative] Third, loops, and the specific reason self-verification underperforms — the part most teams get wrong. [speculative] Fourth, the harness: everything in an agent that is not the model. [speculative] Fifth, the part you came for — what to adopt, defer and refuse on Monday. [speculative] Sections two to four are built from three research reports I will name as I go, so you can check my work rather than trust my summary. [speculative]

# 4 4. What this talk is — and is not

**On screen:**

> *Scope*
>
> ***Covered — around the model***
>
> - *Grounding, verification loops, where the oracle sits [speculative]*
> - *Harness layers: permissions, memory, evaluation, guardrails [speculative]*
> - *Bounded autonomy; what to adopt, defer or refuse [speculative]*
>
> ***Not covered — the model***
>
> - *Prompt tricks, model internals, training, fine-tuning [speculative]*
> - *Which model is "best" this month [speculative]*
> - *Basics you already own: CI, review, testing, containers [speculative]*

A quick fence around the hour, so we do not drift in Q&A. [speculative] On the left, what I will cover: grounding and where the verifying oracle sits, the harness layers, and bounded autonomy — ending each section on an adopt-defer-refuse call. [speculative] Not prompt tricks or model internals; that is the smallest layer. [speculative] Not which model is best this month, because the answer expires before you reach the office. [speculative] Not the delivery basics you already own — CI, review, testing, containers — because you would rightly resent it. [speculative] If you came for "which model should we standardise on," that is the one question this talk deliberately ignores, and section five says why that is the right thing to ignore. [speculative]

# 5 5. Takeaways

**On screen:**

> *Where this goes*
>
> *1*
>
> *Relocate the truth*
>
> *Reliability is bought by moving truth-determination out of the probabilistic model. [S6]*
>
> *2*
>
> *Independence, not iteration*
>
> *A loop's verdict is worth only its verifier's independence, not how hard it iterates. [S10]*
>
> *3*
>
> *Determinism first*
>
> *Push determinism to the load-bearing control; bound the model inside it. [S16]*
>
> *4*
>
> *Autonomy last*
>
> *Unbounded autonomy is unbounded risk [S7], so autonomy is the property you add last. [inference]*

Four takeaways up front, then I earn them for fifty minutes and repeat them at the end — so if you decide you disagree early, you spend the hour building your rebuttal. [speculative] One: reliability is bought by relocating truth-determination out of the probabilistic model. [S6] Two: a loop's verdict is worth what its verifier's independence is worth, not how many times it iterates — independence is the variable. [S10] Three: push determinism to the load-bearing control and bound the model inside it. [S16] Four: because unbounded autonomy in a complex system is unbounded risk [S7], autonomy is the property you add last, after the check is in place — not the feature you lead with. [inference] Hold those four as the destination; everything between here and the end is evidence for one of them or an honest caveat against it. [inference]

# 6 6. Prompt engineering

**On screen:**

> *Part I*
>
> *The centre of gravity moved from the prompt to the system.*

Part one — a signpost, no claim. [speculative] The argument is one move: the centre of gravity shifted from the prompt to the system around it, and — the part that matters for a sceptical room — that shift is now backed by results, not fashion. [speculative] I want to make it carefully, because "prompt engineering is dead" is an overstatement you would be right to reject, and it is not what I am claiming. [speculative] What I am claiming is narrower and harder to argue with. [speculative]

# 7 7. Prompt engineering got demoted

**On screen:**

> *Prompts*
>
> *It did not die — it became the innermost layer of a larger discipline. [S1]*
>
> - *Prompt engineering "demoted itself to the innermost layer of context engineering." [S1]*
> - *The 2025–2026 frontier is negative results: "always add reasoning steps" is now a per-model empirical question, not a universal trick. [S1]*
> - *If your reliability plan is a cleverer prompt, you are optimising the smallest layer. [inference]*
>
> *[S1] Building AI Systems: 5 Pillars — vault research report.*

Prompt engineering did not disappear; it got demoted. [S1] In the words of the five-pillars report this section is built on, it "demoted itself to the innermost layer of context engineering" — still necessary, no longer the outer game. [S1] Here is the tell that the field matured: the 2025-to-2026 frontier is dominated by negative results. [S1] The advice everyone internalised — add chain-of-thought, give it an expert persona — turns out to hurt some reasoning models, so whether to do it is now a per-model empirical question you measure, not a universal trick you assume. [S1] So if your entire plan for making an AI feature reliable is a cleverer prompt, you are pouring effort into the smallest, least durable layer. [inference]

# 8 8. Context is a budget, not a bucket

**On screen:**

> *Context*
>
> *Its maturity is measured by whether context assembly is a deterministic, testable pipeline. [S2]*
>
> - *Context engineering "treats the window as a capped per-turn budget to allocate — not a bucket to fill." [S2]*
> - *The tell for a mature system: you can unit-test what the model was shown. [S2]*
> - *That is already an engineering property, not a prompting one. [inference]*
>
> *[S2] Building AI Systems: 5 Pillars — vault research report.*

The layer outside the prompt is context engineering, with one central discipline. [inference] The report frames it exactly: it "treats the window as a capped per-turn budget to allocate — not a bucket to fill." [S2] The tell that separates a mature context system from an improvised one is testability: you can unit-test what the model was shown, because the assembly is a deterministic, inspectable pipeline, not string concatenation you cannot reproduce. [S2] That is the thesis in miniature: reliability appears the moment you treat the thing around the model as a system with testable properties. [inference]

# 9 9. Why "just add more context" backfires

**On screen:**

> *Context*
>
> *More context is often less reliability — measured, not asserted. [S3]*
>
> *18*
>
> *LLMs where accuracy degrades as length grows, even with irrelevant tokens masked [S3]*
>
> *~4×*
>
> *rise in agentic failure at doubled task duration [S3]*

> *[S3] Building AI Systems: 5 Pillars — vault research report (context rot).*

This is the measurement behind the last slide, and the load-bearing caveat of the context pillar. [inference] The phenomenon has a name — context rot. [S3] Accuracy degrades as the window grows, and it degrades even when the extra tokens are irrelevant and masked — so this is not the model distracted by bad content, it is length itself costing you, observed across eighteen models. [S3] The second number should worry anyone building agents: agentic failure rises roughly fourfold when you double the task duration. [S3] The engineering response is the budget — allocate the window deliberately, and treat "add more" as a cost, not a free win. [inference]

# 10 10. The reframe

**On screen:**

> *Prompts*
>
> *The model is a commodity; the system around it is where reliability lives. [inference]*
>
> - *Long-running-agent failures "are not solved by a smarter model. They are solved by infrastructure." [S24]*
> - *Two teams, same frontier model, wildly different production reliability — the difference is the harness. [S24]*
> - *So the question stops being what do I type and becomes what do I build. [inference]*
>
> *[S24] Harness Engineering — vault concept note.*

Put the last three slides together and the centre of gravity has visibly moved. [inference] The vault's standing note states the sharp version: the recurring failures of long-running agents "are not solved by a smarter model. They are solved by infrastructure." [S24] Make it concrete with a pattern you have lived: two teams take the same frontier model — same weights, same API — and ship products with wildly different reliability. [S24] The delta is not the model, because the model is identical; it is everything they wrapped around it. [S24] That is what I mean by the model becoming a commodity: it is the interchangeable part, and the durable engineering is the system around it. [inference] So the question that decides whether your feature is reliable stops being "what do I type" and becomes "what do I build." [inference] Everything from here answers that second question. [inference]

# 11 11. The five pillars

**On screen:**

> *Part II*
>
> *Five layers, one law running through all of them.*

Part two — the map, no claim on the divider. [speculative] I will lay out five pillars, but do not hear them as a checklist of separate topics. [speculative] They are five layers with one law running through all of them; the law is the thing to remember, the names are just where it shows up. [speculative] I spend the most time on the third pillar, grounding, because that is where the single strongest number in the talk lives and where the thesis is most directly carried. [speculative]

# 12 12. Five pillars, one spine

**On screen:**

> *Pillars*
>
> *[code]*
>
> *Not five topics — one spine, clearest in the grounding pillar. [inference]*
>
> *[S1][S2][S4][S7] Building AI Systems: 5 Pillars — vault research report.*

Here are the five, drawn as a spine rather than a list, because the arrow between them is the point. [inference] Prompt is the innermost layer — covered. [S1] Context is the budget around it — covered. [S2] Grounding, in green, is where we are headed and the heart of the thesis: condition on evidence, then verify against something external. [S4] Orchestration is how you compose agents, and its load-bearing move is splitting the agent that does the work from the agent that checks it. [[Concepts/AI Agent Infrastructure/Loop Engineering]] Bounded autonomy is the governance layer that wraps all of it. [S7] The reason I draw one spine is that the same discipline recurs at every joint: you move the guarantee off the probabilistic component onto something you can inspect. [inference] If you take one pillar home, take grounding — so let me go there and slow down. [inference]

# 13 13. Grounding, in three layers

**On screen:**

> *Grounding*
>
> *Reliability comes from closing a generation–verification loop against an external oracle. [S4]*

> *[code]*
>
> *Grounding "decomposes into parametric correlation, structural grounding, and deterministic verification." [S4]*
>
> *[S4] Building AI Systems: 5 Pillars — vault research report.*

Grounding is not one thing, and conflating its layers is where a lot of production RAG quietly goes wrong. [inference] The report decomposes it into three epistemically distinct layers. [S4] Layer one is parametric correlation — the model answering from its own weights, its memory; a correlation, not a guarantee. [S4] Layer two is structural grounding: you condition on retrieved evidence, which narrows what it can plausibly say. [S4] Layer three is deterministic verification, where the answer is decided by something that is not the model — a test, a solver, a schema — and the model is demoted to orchestrator. [S4] Reliability, the report says, comes from closing that generation-and-verification loop against an external oracle — not from a more capable model at layer one. [S4]

# 14 14. Only the deterministic layer certifies

**On screen:**

> *Grounding*
>
> *Retrieval narrows the output space; only a deterministic check gives a causal guarantee. [S41]*
>
> - *Structural (retrieval) grounding improves the odds — it does not certify anything. [S41]*
> - *Only deterministic verification, where the model is not the load-bearing component, "approaches a genuine causal guarantee (Layer 3)." [S41]*
> - *Match the layer to the stakes: an advisory answer rides on retrieval; an irreversible action — a deploy, a migration, a payment — needs a deterministic gate. [inference]*
>
> *[S41] Structural Grounding — vault concept note (three-layer grounding taxonomy).*

The line between layer two and layer three is the one that tells you where to put your effort. [inference] Retrieval grounding — layer two — improves your odds; it conditions on real evidence and narrows the output space, and it is worth doing. [[Concepts/LLM/Structural Grounding]] Only the deterministic layer — where the model is explicitly not load-bearing — approaches a genuine causal guarantee. [[Concepts/LLM/Structural Grounding]] Match the layer to the stakes of the action. [inference] But the moment the action is irreversible — a deploy, a migration, a payment, a delete — you want a deterministic check between the model and the effect, not a more confident model and a hope. [inference]

# 15 15. The number that should change your mind

**On screen:**

> *Grounding*
>
> *The same agents, open loop → closed loop, model-invariant across DeepSeek and Claude. [S5]*
>
> *56.8%*
>
> *compliance — open loop, no external verifier [S5]*
>
> *98.6%*
>
> *compliance — closed loop, external FE verifier [S5]*
>
> *Nothing about the model changed; the generation–verification loop did. [S5] But scope bounds the guarantee: a loop fixes only the errors its verifier can see — a defect outside its scope passes through. [S37]*
>
> *[S5] Building AI Systems: 5 Pillars — vault research report (closed-loop external-verifier study). [S37] 5 Pillars — Grounding, the study's own boundary condition — vault research report.*

This is the strongest single number in the talk, so I will slow down and be precise — precision is what this room will demand. [inference] A published 2026 study ran a set of agents open-loop, no external verifier, and measured 56.8% compliance. [S5] Holding the agents the same, it closed the loop with an external FE — finite-element — verifier and measured 98.6%. [S5] Same agents; the only change is that a check the model could not grant itself now sat in the loop, and the gain held across two model families, DeepSeek and Claude. [S5] It is a safety-critical engineering-compliance benchmark — a domain that owns a clean external verifier, and one that is not yours; I cite it purely as the cleanest external-verifier result I know, not as your example. [inference] What transfers is the mechanism, not the domain — but the mechanism is not unbounded, and the study's authors flag the limit honestly, so I will too. [S37] A verification loop fixes only the errors its verifier can see: in their own case, when a defect came from structural topology rather than the parameters the FE verifier checks, the closed loop was "powerless" to catch it. [S37] So the guarantee generalises across domains, but never past the scope of the oracle you build — I return to that on the honest-limits slide. [S37] Let the number sit before I say why it is model-invariant. [inference]

# 16 16. What the number says

**On screen:**

> *Grounding*

> *The mechanism matters more than the magnitude. [inference]*
>
> - *The gain is model-invariant because "the guarantee now lives in the verifier, not the generator." [S15]*
> - *The general pattern: "reliability is bought by relocating the truth-determining computation out of the probabilistic model." [S6]*
> - *Architecture beats scale. [inference]*
>
> *[S15] Loop Engineering · [S6] 5 Pillars — vault research reports.*

Why is that gain model-invariant — why does it survive swapping DeepSeek for Claude? [inference] The Loop Engineering report gives the reason in one line: because "the guarantee now lives in the verifier, not the generator." [S15] Once the thing that certifies correctness is external, changing the generator does not move the guarantee. [S15] That is the general pattern behind every pillar, stated by the five-pillars report as plainly as I can put it: "reliability is bought by relocating the truth-determining computation out of the probabilistic model." [S6] The reflex when a feature is unreliable is "wait for the smarter model" or "scale it up." [inference] This says the leverage is elsewhere: architecture beats scale, because architecture decides where truth gets determined — and a bigger model at layer one is still at layer one. [inference]

# 17 17. What is an oracle?

**On screen:**

> *Grounding · definition*
>
> *An oracle is a check that can tell right from wrong without asking the model. [inference]*
>
> - *A test's exit code, a type checker, a solver's verdict, a schema validator — the verdict comes from somewhere the generator does not control. [inference]*
> - *The defining property is negative: the model cannot satisfy it by asserting it is done. [inference]*
> - *It is what reliability is bought with — "closing a generation–verification loop against an external oracle where the LLM is orchestrator, not load-bearer." [S4]*
>
> *[S4] 5 Pillars — grounding, three layers — vault research report.*

One word has been carrying a lot of weight, so let me define it before I lean on it further. [inference] An oracle, in this talk, is a check that can tell right from wrong without asking the model — a test's exit code, a type checker, a solver's verdict, a schema validator, a business-rule engine. [inference] The defining property is negative: the generator cannot satisfy it by asserting it is done, because the verdict comes from somewhere the model does not control. [inference] That is exactly the thing the five-pillars report says reliability is bought with — closing a generation-and-verification loop against an external oracle, with the model as orchestrator rather than the load-bearing component. [S4] Hold that definition, because the next slide asks the only question that decides your adopt call: for the task in front of you, which oracle do you actually have? [inference]

# 18 18. Which oracle do you actually have?

**On screen:**

> *Grounding · triage*
>
> *The guarantee is bounded by your oracle, not the domain. [S37]*
>
> *1*
>
> *Oracle-rich*
>
> *A cheap external check exists — schema validator, contract test, migration dry-run. Close the loop, get the guarantee. [inference]*
>
> *2*
>
> *Oracle-scoped*
>
> *The verifier sees only part — a type checker proves types, not logic. It certifies only what it checks. [S37]*
>
> *3*
>
> *Oracle-free*
>
> *Is this refactor correct? This summary good? No complete oracle — partial checks, bounded review, sometimes refuse. [inference]*
>
> *[S37] 5 Pillars — Grounding boundary — vault report.*

Now the question that is the hinge of your adopt decision, and the objection this room is already forming: "that study has a finite-element verifier — I don't have one for a pull request." [inference] Correct — so sort the task into one of three zones, each with an example from your own world. [inference] Oracle-rich: a cheap external check already exists — a schema validator on an API response, a contract test on a service boundary, a migration dry-run — close the loop and get something close to the flagship guarantee. [inference] Oracle-scoped: a verifier sees only part of the space — a type checker proves your types are sound but says nothing about the business logic — so the loop certifies what it checks and no more; the flagship study's own FE verifier is parameter-level, and a defect from structural topology passed straight through. [S37] Oracle-free: most open-ended work — is this refactor correct, is this summary good — has no cheap oracle at all, and there external grounding does not save you; the honest architecture is bounded autonomy plus human review, and sometimes the call is to refuse. [inference] I won't pretend the oracle is free — building it is the real cost, which I return to on the honest-limits slide. [inference] The move is diagnostic: decide which zone the task is in, because that decision is the adopt-defer-refuse call itself. [inference]

# 19 19. Cheap checks you already own

**On screen:**

> *Grounding · oracle-free zone*
>
> *No complete oracle rarely means no check. [speculative]*
>
> - ***Property-based tests** — assert invariants, not exact outputs. A sound one synthesizes "in 2.4 samples on average", but covers only 21% of properties. [S38]*
> - ***Metamorphic relations** — equivalent inputs must not change the output. On HumanEval, caught "75% of the erroneous programs generated by GPT-4" at an 8.6% false-positive rate. [S39]*
> - ***Canary / shadow** — route 1% of traffic, alarm on the error-rate delta. Production is the oracle. [S40]*
> - ***Idempotency** — an idempotent write or reversible migration; a wrong call is safe to retry. [S42]*
>
> *[S38][S39][S40][S42] — vault notes.*

This is the answer to the sharpest objection in the room — "my pull request has no finite-element verifier." [speculative] True, and you rarely have a complete one; but oracle-free almost never means check-free — you already own partial oracles that raise the floor, and each maps onto something you run. [speculative] Property-based testing asserts invariants rather than exact outputs — the workhorse is a round-trip: assert deserialize of serialize of x equals x for random x, catching a class of encoder bugs with no hand-written cases — and — with the best model and prompting approach — a valid, sound one synthesizes in about 2.4 samples on average, though that same best model covers only twenty-one percent of properties, so it is a best-case figure, not a general rate. [S38] Metamorphic testing checks that equivalent inputs give consistent outputs, no ground truth needed — reorder two independent query filters, or rename a variable, and the result must not change — and on the HumanEval benchmark it caught seventy-five percent of the erroneous programs generated by GPT-4, at under a nine-percent false-positive rate. [S39] Canary and shadow deployment make production itself the oracle: route one percent of traffic to the new agent, alarm on the error-rate delta, blast radius you set. [S40] And the oldest tool in the box: make the action idempotent — a write behind a key, or a migration with a real down-step — so executing it once has the same effect as twice, and a wrong call is safe to retry, not a thirteen-hour outage. [S42] None is as clean as an FE solver, but stacked they turn "no oracle" into "good-enough oracle" — usually the honest answer for this room. [speculative]

# 20 20. Split the maker from the checker

**On screen:**

> *Orchestration*
>
> *Independence here is primarily a context boundary. [inference]*
>
> *[code]*
>
> ***Maker** produces the work; **checker** judges it — the deck's one name for this split. The sources name the same two roles generator vs verifier [S15] and observer independence [S9]. [inference]*
>
> *The checker sees the task and the result, never the maker's chain of thought. [S26]*
>
> *[S26] Loop Engineering — vault concept note · [S9] · [S15] — vault research reports.*

Orchestration is the fourth pillar — how you compose more than one agent — and its load-bearing move is one idea: split the agent that makes from the agent that checks. [[Concepts/AI Agent Infrastructure/Loop Engineering]] One vocabulary note so nothing trips you up: I call these two roles the maker and the checker throughout, and the papers I quote name the same split differently — generator versus verifier, or the independence of the observer — but it is the same two boxes every time. [inference] It is primarily a context boundary, not necessarily a different model: the checker sees the task and the result and deliberately not the maker's chain of thought. [[Concepts/AI Agent Infrastructure/Loop Engineering]] A checker who reads the maker's reasoning is biased toward confirming it; strip the reasoning and the check becomes meaningful. [inference] And on the "how many agents" debates: the note is explicit that what carries the effect is the split — maker ≠ checker — not the count, an implementation degree of freedom rather than the load-bearing property. [[Concepts/AI Agent Infrastructure/Loop Engineering]] So ten agents sharing one context, with no real boundary between making and checking, buy you less than two that have one — the split is the point, not the headcount. [inference]

# 21 21. Fewer agents, sharper interfaces

**On screen:**

> *Orchestration*

> *The interface between agents is the design, not the head-count. [inference]*
>
> - *Intuit rebuilt its agent architecture twice in about four months: specialist-agent fleet → central orchestration layer → skills-and-tools. [S27]*
> - ***Second rebuild:** natural-language handoffs meant "a ten agent chain did not fail occasionally, it compounded errors by design." [S27]*
> - ***First rebuild:** [S27] gives no cause for it, so I won't invent one. [inference]*
> - *Endpoint: "skills-and-tools, not agent-to-agent natural-language delegation." A free-text handoff is an untyped interface between unreliable components — you wouldn't ship it between services. [S27]*
>
> *[S27] Intuit Scrapped Its Agent Architecture Twice in Four Months — VentureBeat, 2026. https://venturebeat.com/orchestration/intuit-scrapped-its-own-ai-agent-architecture-twice-in-four-months-at-vb-transform-2026-its-ai-vp-called-that-the-fast-path*

Here is a team that learned the maker-checker corollary the expensive way, in public. [inference] Intuit rebuilt its agent architecture twice in about four months — a specialist-agent fleet, then a central orchestration layer, then a skills-and-tools design. [S27] Be precise about what the source explains, because two rebuilds is two decisions and I only have one reason. [inference] The *second* rebuild is sharp — I'll quote their AI VP because the phrasing is exact: "a ten agent chain did not fail occasionally, it compounded errors by design." [S27] The *first* — why the fleet became an orchestration layer — the source does not give; it reports the endpoint, not that step, and I won't reverse-engineer a motive. [inference] The fix was not a better model or more agents; it was replacing free-text delegation with typed skills and tools: "skills-and-tools, not agent-to-agent natural-language delegation." [S27] Translate to an instinct you have: a natural-language handoff is an untyped, unvalidated interface between two unreliable components — you would never ship that between two of your services, so don't ship it between two agents. [inference]

# 22 22. Freedom inside a fence

**On screen:**

> *Bounded autonomy*
>
> *Autonomy is a calibrated design parameter, not a default. [S43]*
>
> - *"Unbounded autonomy in complex systems inherently creates unbounded risk." [S7]*
> - *Humans keep authority over the boundary, not every step inside. [S43]*
>
> *22.2% → 59.0%*

> *target safety factor hit, oracle off → on (p = 0.0008) [S45]*
>
> *4.2×*
>
> *more structural complexity, still bounded [S45]*
>
> *The fence doesn't shrink the agent's reach — it lets it do more. A CAD study, not your domain. [inference]*
>
> *[S7] · [S43] Bounded autonomy — vault report + note. [S45] Physics-in-the-Loop, IJCAI-ECAI 2026 — 5 Pillars report.*

The fifth pillar is governance, and the core claim is blunt: "unbounded autonomy in complex systems inherently creates unbounded risk." [S7] The vault reframes autonomy as a calibrated design parameter, not a default: freedom inside a fence someone else drew, with humans keeping authority over the boundary conditions rather than approving every step inside it. [[Concepts/Active Inference/Bounded Autonomy in AI]] Now the part that surprises people, because it inverts the intuition that a constraint costs capability — one clean measured result, from a CAD study I flag up front as not your domain, cited only for the mechanism. [inference] Physics-in-the-Loop put a deterministic finite-element solver inside an agent's design loop as the fence, then ran the ablation: with the oracle off, the agents hit the target safety factor 22.2% of the time; with it on, 59.0%, at p = 0.0008 — more than double the hit rate, and they produced designs 4.2 times more complex, not fewer. [S45] The bound is not a leash that shortens the agent's reach; it is the floor that lets it explore aggressively without falling through — and the fence is what makes the freedom safe to grant. [inference] A fence the model can talk its way past is not a fence. [inference]

# 23 23. Loops

**On screen:**

> *Part III*
>
> *The one thing a loop must not do is grade its own work.*

Part three — the section most teams get wrong, so it is worth the time; no claim on the divider. [speculative] The pattern I see everywhere: a team builds a loop where the model produces something, the same model is asked "is this correct?", and if it says yes the loop stops and ships. [speculative] They call that verification. [speculative] The research says self-grading is the one thing a loop must not do, and I will give both the theory and the measurements, because a claim this counterintuitive needs both. [speculative]

# 24 24. Two loop archetypes

**On screen:**

> *Loops*
>
> *What carries the effect is the independence of the observer, not the loop itself. [S9]*
>
> *[code]*
>
> *One loop closes against an external oracle; the other re-invokes the same model. [S9]*
>
> *[S9] Loop Engineering — two loop archetypes — vault research report.*

There are two shapes of loop, and on the surface they look almost identical — both iterate, both have a pass-or-retry decision. [inference] The top loop closes against an external oracle: a test, a solver, an independent judge decides pass or fail, and only a real pass exits. [[Concepts/AI Agent Infrastructure/Loop Engineering]] But the finding that reorganises this field is that "what carries the effect is the independence of the observer, not just the presence of a loop." [S9] Iteration count is not the variable; independence is. [inference]

# 25 25. Independence, not iteration

**On screen:**

> *Loops*
>
> *One task, three checkers — audit an agent's work for real violations. Only independence differs. [S47]*
>
> *0 / 34*
>
> *iterated self-check: the agent audits itself [S47]*
>
> *7 / 34*
>
> *weak independence: fresh instance, same model [S47]*
>
> *34 / 34*
>
> *external oracle: a deterministic checker [S47]*
>
> > *"A loop's verdict is only worth what its verifier's independence is worth." [S10]*

> *Iterating a self-graded loop just gets a confident wrong answer faster. [inference]*
>
> *[S10] Loop Engineering — verifier independence · [S47] Loop Engineering — self-audit study — vault reports.*

Here is the whole section on one concrete task, because a number lands harder than a principle. [inference] A controlled study ran the same job — audit an agent's work for real violations — three ways. [S47] Iterated self-check: the agent audits its own output and flags zero of thirty-four violations, at ninety-to-a-hundred confidence — a confident miss. [S47] Weak independence: a *fresh* instance of the same model, shown only the work and not the reasoning, catches seven — better, because the context is isolated, though still the same weights. [S47] External oracle: a deterministic checker catches all thirty-four. [S47] The model is identical in the first two; what changed is how independent the check's evidence is. [inference] So the sentence to keep: "a loop's verdict is only worth what its verifier's independence is worth; buy that independence in the check, or don't call the loop verified." [S10] You buy it in the check — an isolated context, a deterministic gate, a different family for subjective cases — or you do not get to say "verified," however many times it iterated. [S10] Iterating a self-graded loop harder just gets you the confident wrong answer faster. [inference]

# 26 26. Why self-check hits a wall

**On screen:**

> *Loops*
>
> *Self-check is bounded structurally, not by capability. [inference]*
>
> - *Gödel showed "a formal system rich enough to describe arithmetic cannot prove its own consistency from inside." [S46]*
> - *As an analogy: a same-system checker has no outside vantage point — "the AI self-check wall is Gödel with a leaderboard." [S11]*
> - *Not "self-checking is impossible" — "trustworthy knowledge always has to pay for its own verification; the price is never zero." [S11]*
> - *Asked to verify, an LLM re-runs the same process; the fix is a paid independent channel, not a smarter grader. [inference]*
>
> *[S11] Loop Engineering — narrowed claim · [S46] — the Gödel step — vault reports.*

Why does self-check hit a wall — is it just that the models are not smart enough yet? [inference] Here is the mechanism, the argument to lead with: a self-monitor grades its own output using the same information that produced it, so re-invoking the same weights opens no independent channel — extra rounds redistribute confidence, they do not add evidence. [inference] The report has a memorable label — "the AI self-check wall is Gödel with a leaderboard" — and I use it because it sticks, but a name is not an explanation, so let me actually say what Gödel showed. [S11] He proved that a formal system rich enough to describe arithmetic cannot prove its own consistency from inside itself. [S46] I am borrowing that as an analogy, not smuggling in a theorem about neural networks — the only honest transfer is this: a checker built from the same system as the thing it checks has no vantage point outside it, which is why re-invoking the same weights adds no evidence. [inference] But here is where I stay honest, because this is exactly the spot where the field over-claims: the report explicitly refuses to inflate that label into "self-checking is impossible," and neither will I. [S11] The version it actually stands behind is narrower and more useful — "trustworthy knowledge always has to pay for its own verification; the price is never zero." [S11] Which points straight at the fix: not a smarter grader, but a different, independent channel of evidence — and you pay for it every time. [inference]

# 27 27. The empirical case · 1

**On screen:**

> *Loops*
>
> *On-policy self-monitoring — a model grading a patch it just wrote — measurably loses discrimination. [S12]*
>
> *0.99 → 0.89*
>
> *monitor AUROC grading its own patch (1.0 = perfect) [S12]*
>
> *5×*
>
> *in one setting: likelier to wave through a prompt-injected patch [S12]*
>
> *One 2026 preprint, four coding/tool-use datasets — self-attribution bias: the model defends its own work. [S12] Read the direction, not the multiple — directionally robust, not settled. [S12]*
>
> *[S12] Loop Engineering — empirical case — vault research report (single 2026 preprint, Khullar et al., arXiv:2603.04582).*

So much for theory — now the measurements, and I will be deliberately careful about scope, because this room will check and this deck may be published. [inference] This is one 2026 preprint — Khullar and colleagues — measuring what they call on-policy self-monitoring, which just means the model grading output from its own policy: it checks a patch it wrote itself. [S12] First finding: when the model grades that patch, its discrimination degrades. [S12] The metric is AUROC — read it as a right-versus-wrong separation score, where 1.0 is perfect separation and 0.5 is a coin toss — and it drops from 0.99 to 0.89 in the code-correctness setting. [S12] Against that scale 0.99 is near-perfect and 0.89 is a small but real loss you can check for yourself, and it appears only because the work is the model's own. [inference] Second finding, and let me fence it precisely: in one setting of that study, self-attribution bias made the monitor five times likelier to wave through a patch that had followed a prompt injection. [S12] I flag the scope on purpose — it is a single preprint, and the report's own Skeptic reading calls these magnitudes "directionally robust but not settled," so do not walk out repeating "5×" as a law of nature. [S12] What the design rests on is the direction, not the multiple: a self-monitor is weakest exactly where authorship bias bites hardest, which is under attack. [inference]

# 28 28. The empirical case · 2

**On screen:**

> *Loops*
>
> - *On formal-verification benchmarks: "rule-based approaches remain more reliable for verification success, while LLM-based methods exhibit more variable performance." [S13]*
> - *The paper's own framing is the careful one — not "rules beat LLMs," but rules more reliable, LLMs more variable and complementary. [S13]*
> - *Practitioner reading: default the check to a deterministic gate — an exit code, a schema, a solver — and reach for a model judge only where the target is irreducibly subjective. [inference]*
>
> *[S13] Loop Engineering — empirical case — vault research report.*

The second measurement comes from formal verification, and I quote it precisely because it is easy to overstate and I have overstated it before. [inference] The study — LLM-generated formal annotations checked against a verifier and SMT solvers — found that "rule-based approaches remain more reliable for verification success, while LLM-based methods exhibit more variable performance." [S13] It positions the LLMs as complementary tools that are more variable, not simply worse — the deterministic route is the reliable one, the model the variable one. [S13] Reach for a model judge only where the target is irreducibly subjective. [inference] It is a priority ordering, not a prohibition. [inference]

# 29 29. This is consensus, not one author

**On screen:**

> *Loops*

> *Two-loop report*
>
> *[S9]*
>
> *Independence of the observer carries the effect. [S9]*
>
> *Theory*
>
> *Practitioner rule*
>
> *[S14]*
>
> *Never close a loop against the model that opened it. [S14]*
>
> *Design rule*
>
> *Security result*
>
> *[S17]*
>
> *Deterministic control beats prompt-level self-defense; 0% attack success with hand-written policies. [S17]*
>
> *Field result*
>
> *Grounding result*
>
> *[S5]*
>
> *External verifier lifts 56.8% → 98.6%, model-invariant. [S5]*
>
> *Field result*

Before the design rule, let me defend the word "consensus," because a sceptic reading the footnote column has every right to ask whether this is one report agreeing with itself. [inference] The theory — independence of the observer carries the effect — comes from the Loop Engineering report. [S9] The design rule — never close a loop against the model that opened it — is that report's practitioner synthesis. [S14] The security result is Progent, an external SMT-verified privilege control where deterministic control beats prompt-level self-defense and drives attack success to zero with hand-written policies — a different group, domain and benchmark. [S17] And the grounding result — the external verifier lifting the same agents from 56.8 to 98.6 percent — comes from the five-pillars work and rests on a separate published study again. [S5] Four facets, three independent strands, one conclusion: put the verifier outside the model. [inference]

# 30 30. So what do you build

**On screen:**

> *Loops*
>
> *The whole section collapses to one rule. [inference]*
>
> *[code]*
>
> > *"Never let a loop close against the same model that opened it; close it against an external oracle, or don't call it verification." [S14]*
>
> *[S14] Loop Engineering — practitioner synthesis — vault research report.*

The whole section collapses to one implementable rule: "never let a loop close against the same model that opened it; close it against an external oracle, or don't call it verification." [S14] The checker runs in an isolated context — it sees goal and result, never the maker's reasoning — and cannot edit the gate it is judged against. [[Concepts/AI Agent Infrastructure/Loop Engineering]] And the retry arrow is bounded: two circuit breakers, an iteration ceiling and a token budget, stop the loop running forever or burning your account when it cannot converge. [[Concepts/AI Agent Infrastructure/Loop Engineering]] One sequencing note into the next section: add the checker before you add the autonomy, because the check is what makes the autonomy safe to grant. [inference]

# 31 31. The harness

**On screen:**

> *Part IV*
>
> *One law runs through every layer of it.*

Part four — the harness — no claim on the divider. [speculative] Let me define the term once, in the flow, because it is the one piece of vocabulary this talk rests on. [speculative] The harness is everything in an agent that is not the model: the tools, sandboxes, memory, sensors, permissions, orchestration — the whole control plane the model runs inside. [speculative] This section shows the one law that runs through every layer, walks a single layer all the way down, and then spends real time on the caveats — because a section that only sold you the harness would be dishonest, and you would know it. [speculative]

# 32 32. Model held constant, harness changed

**On screen:**

> *Harness*
>
> *At the frontier, harness work pays back faster than a model swap. [speculative]*
>
> *~1M*
>
> *lines shipped via ~1,500 automated PRs, zero hand-written (OpenAI) [S25][S34]*
>
> *+13.7*
>
> *Terminal Bench 52.8 → 66.5, harness changes only, model held constant (LangChain) [S25] [S35]*
>
> *[S34] Harness engineering — OpenAI. https://openai.com/index/harness-engineering/ [S35] Improving deep agents with harness engineering — LangChain.*

Two anchors, both first-party, both chosen because they hold the model constant and move only the harness — the controlled comparison you would ask for. [inference] OpenAI's team reported shipping roughly a million lines of production code through about 1,500 automated pull requests with zero hand-written code — a harness achievement, not a model release. [S25] [S34] The cleaner controlled experiment is LangChain's: they lifted a coding agent from 52.8 to 66.5 on Terminal Bench — 13.7 points — purely by changing the harness, model held fixed. [S25] [S35] Same weights, thirteen points, from the scaffolding alone. [inference] The reading is a resource-allocation one, and it is a live decision for your teams: at the current frontier, harness work pays back faster than waiting for a better model. [speculative] The model you already have almost certainly has more headroom in its harness than in its weights. [speculative]

# 33 33. The one law

**On screen:**

> *Harness*
>
> > *"Push determinism to the load-bearing control, and bound the probabilistic model inside it." [S16]*
>
> *[code]*
>
> *The same law recurs at every layer of the harness. [S16]*
>
> *[S16] Harness Engineering — vault research report.*

This is the organising law of the whole talk, the engine under the four takeaways: "push determinism to the load-bearing control, and bound the probabilistic model inside it." [S16] Read it slowly. [inference] The load-bearing control — the thing that must not fail — is deterministic. [inference] The model is bounded inside that control, doing the flexible part where being occasionally wrong is survivable. [inference] You are not trying to make the model reliable; you are arranging for the reliable part to be the part that carries the weight. [inference] And the report's claim, which the next slides make good on, is that this same move recurs at every layer of the harness. [S16] Every example I show from here is one instance of this one sentence — if I make a claim that does not reduce to it, call it out. [inference]

# 34 34. Five layers, one law

**On screen:**

> *Harness*
>
> *[code]*
>
> *At each layer: determinism at the load-bearing control, the model bounded inside it. [S16] [S24]*
>
> *[S16] Harness Engineering — the one law — vault research report · [S24] Harness Engineering — vault concept note.*

Here is the control plane as five layers wrapping the model. [[Concepts/AI Agent Infrastructure/Harness Engineering]] Permissions and sandboxing decide what the agent may do. [inference] Memory decides what persists across turns. [inference] Evaluation and observability decide whether you can see and measure what it did. [inference] Orchestration decides how agents compose, and runtime guardrails decide what gets stopped at the moment of action. [inference] What makes this one discipline rather than five unrelated concerns is that the same law holds at each: push determinism to the load-bearing control and bound the model inside it. [S16] I will not march through all five and bore you — instead I take the first layer, permissions, and walk it all the way down, because it has the cleanest quantified evidence and the sharpest caveat. [inference]

# 35 35. Determinism at the load-bearing control

**On screen:**

> *Harness · permissions*

> *Determinism at the control does what a defensive prompt cannot. [inference]*
>
> - *Progent — a deterministic privilege control verified with an **SMT** (Satisfiability Modulo Theories) solver — "provably reduces attack success rate to 0% with hand-written policies, while prompt-level defenses are demonstrably bypassed." [S17]*
> - *The provable part is the mechanism: SMT-verified monotonic confinement — the engine provably enforces whatever policy it is given. [S17]*
> - *The 0% is a measured benchmark result — on AgentDojo and **ASB** (Agent Security Bench), under hand-written policies — not itself a theorem. [S18][S32]*
>
> *[S17] Harness Engineering — vault report · [S32] Progent — arXiv 2504.11703. [https://arxiv.org/abs/2504.11703](https://arxiv.org/abs/2504.11703)*

Take permissions as the worked example. [inference] With hand-written policies Progent "provably reduces attack success rate to 0%, while prompt-level defenses are demonstrably bypassed." [S17] Two bits of vocabulary this room will want exact: SMT is Satisfiability Modulo Theories — the solver technology that lets you prove a policy has a property — and ASB is the Agent Security Bench. [inference] Now be precise about what "provable" attaches to, because it does two jobs and only one is a proof. [inference] The SMT verification proves the *mechanism*: the policy engine provably enforces whatever policy it is handed — the paper calls this monotonic confinement. [S17] The *0% figure* is not proven; it is *measured* — that mechanism, evaluated on the AgentDojo and ASB benchmarks under hand-written policies, drove attack success to zero. [S32] Proven mechanism, measured result — not the same claim, and this room would catch me if I let "provable" cover both. [inference] So determinism at the load-bearing control does what a defensive prompt structurally cannot — but I won't leave you with only the 0%, because the next slide is where the honesty lives. [inference]

# 36 36. But: who writes the policy?

**On screen:**

> *Harness · permissions*
>
> *Hand-written policy*
>
> *0% attack success — measured on the benchmarks, not a theorem. [S18]*
>
> *LLM-generated policy*
>
> *Residual attack survives: AgentDojo 41.2% → 2.2%, ASB 70.3% → 7.3%. [S18]*

> *Even deterministic control degrades once the model authors its own policy. [S18] A privilege policy attaches to a tool-class, not a task, so the hand-written slice amortises across every task that uses that tool. [inference]*
>
> *[S18] Harness Engineering — vault research report (Progent, autonomous-policy config).*

Here is the nuance I refuse to gloss to "near-zero," because glossing it would be the exact over-claim this talk is against. [inference] And carry the precision from the last slide with you: that 0% is a *measured*, hand-written-policy result — the confinement mechanism is what's proven, the zero is what was observed on the benchmarks. [S18] The moment you let the model author its own policy — which is what everyone wants, because writing policies by hand does not scale — the residual attack comes back: AgentDojo falls from 41.2% to 2.2%, ASB from 70.3% to 7.3%. [S18] Big reductions, single digits — but single digits are not zero, and the gap between zero and 2.2% is the gap between a guarantee and a good average. [inference] Read the pattern precisely: even a deterministic control degrades once the probabilistic model is the thing writing the deterministic rules — you have quietly moved the model back onto the load-bearing path. [S18] So the division of labour on this slide: the human owns the policy, the machine executes it. [inference]

# 37 37. The harness is an attack surface

**On screen:**

> *Harness*
>
> *Prompt-layer defenses fail; execution-layer controls are required. [S19]*
>
> *100+*
>
> *coding agents compromised via poisoned MCP tools / indirect prompt injection [S19][S29]*
>
> *Corroborated across outlets: 100+ agents, Fortune 500 and Fortune 100. [S29]*
>
> *[S19] Harness Engineering — vault report · [S29] Agentjacking — vault watch note.*

There is a second reason the harness matters, less comfortable than the reliability one: it is where you get attacked. [inference] Agentjacking is the name — attackers compromised over a hundred coding agents through poisoned MCP tools and fake bug reports, feeding malicious instructions in through the tool-and-data channel the agent trusts. [S19] That is the empirical case that prompt-layer defenses fail and execution-layer controls are required — you cannot prompt your way out of a poisoned tool result, because the poison arrives as data the model is supposed to act on. [S19] The 100-plus figure is corroborated across multiple outlets and hit Fortune 500 and Fortune 100 companies — not a lab curiosity. [S29] Everything you know about not trusting input at a boundary applies, at the execution layer, not the prompt layer. [inference]

# 38 38. Cite the number you can defend

**On screen:**

> *Harness · evidence discipline*
>
> *Cite this*
>
> *100+ agents compromised — corroborated across outlets. [S29]*
>
> *Not this*
>
> *85% success rate — "remains single-source (Tenet Security) as of July 2026." [S30]*
>
> *On a public slide, cite the number you can stand behind, not the loudest one. [inference]*
>
> *[S29] Agentjacking · [S30] 85% not independently replicated — vault watch notes.*

Let me step out for one slide and show the discipline behind the argument, because it is a move I want you to be able to make. [inference] When I researched agentjacking, the eye-catching figure was an "85% success rate" — a great number for a slide, and single-source, from one vendor, Tenet Security, not independently replicated as of July 2026. [S30] The defensible figure is the boring one: "100-plus agents compromised," which multiple independent outlets corroborated. [S29] On any slide you will be challenged on, cite the number you can stand behind, not the one that gets the gasp — and that is the third Monday move, done live. [inference]

# 39 39. Not every harness change helps

**On screen:**

> *Harness*
>
> *The harness is leverage in both directions. [inference]*
>
> *93.1% → 55.2%*
>
> *Codex retrieval accuracy: inline → file-based grep delivery [S28]*
>
> - *Same model; the only change was how tool output reached it — inline vs written to a file. ~38 points. [S28]*

> - *Authors' read: file-based routing "trades context bandwidth for compositional tool competence; gains are realized only when the agent reliably closes the loop." [S28]*
> - *The paper gives no full error analysis for the collapse — so take the direction as the lesson, not a mechanism. [inference]*
>
> *[S28] Is Grep All You Need? — vault reading note.*

The honest caveat to "just engineer the harness and reliability follows": the harness is leverage in both directions. [inference] In one measured case, swapping a single tool-delivery format — inline results versus file-based grep — dropped Codex from 93.1% to 55.2%, roughly thirty-eight points on the same model, from one seemingly-cosmetic choice about how a tool returns output. [S28] The authors' mental model is worth borrowing: they read file-based routing as trading context bandwidth for compositional tool competence, where "gains are realized only when the agent reliably closes the loop" — writing results to a file pays off only if the agent then reliably issues the follow-up calls. [S28] Whether that is precisely what Codex failed to do, the paper does not say — it reports the collapse without a full error analysis, so on the note's own reading the finding is descriptive rather than explanatory. [S28] I won't oversell it, then: the honest takeaway is the direction — a cosmetic-looking choice can swing accuracy by forty points — not a validated causal story. [inference] Which means the harness discipline and the evaluation discipline are one: you do not know whether a harness change helped until you re-run the eval set. [inference]

# 40 40. Harnesses decay — delete on a cadence

**On screen:**

> *Harness*
>
> - *A scaffold built for a model's weakness becomes overhead — and can cap capability — once that weakness disappears. [S44]*
> - *The cadence the source supports is model-driven, not calendar-driven: "re-run the harness against your eval set on every model bump" — one cited blog approximates that as quarterly. [S44]*
> - *It concerns the human-authored scaffolding teams accumulate; the source does not scope this to AI-generated harnesses. [inference]*
> - *The audit each time: remove a component, re-run the eval set, keep only if quality holds — otherwise revert. [S44]*
>
> *[S44] Harness Engineering — vault concept note (deletion cadence).*

One more caveat, so none of this becomes cargo cult. [inference] A harness component built to compensate for a model's weakness becomes pure overhead the moment that weakness disappears — and worse, it can actively cap the model's current capability, holding a better model down to the shape of the worse one. [S44] Scaffolding accretes; nobody is incentivised to delete it; two generations later you are running a Rube Goldberg machine of workarounds for problems that no longer exist. [inference] So be precise about "what cadence": the source ties it to model upgrades, not the calendar — "re-run the harness against your eval set on every model bump," which its one cited blog approximates as roughly quarterly, a proxy for model-release pace, not a fixed interval. [S44] And whose harness: the human-authored scaffolding teams accumulate — the source makes no claim about AI-generated harnesses, so neither will I. [inference] The ritual each time: remove a component, re-run the eval set, keep it only if quality holds — otherwise revert. [S44] Your harness should get smaller on some upgrades, not only larger. [inference]

# 41 41. When the fence is missing: Amazon Kiro

**On screen:**

> *Harness · Kiro incident*
>
> *A permission vacuum, not a weak model: the agent took the most destructive path because nothing stopped it. [S20]*
>
> *No deterministic gate*
>
> *Kiro "autonomously decided to delete and rebuild the production environment" — Cost Explorer, one region, ~13-hour outage. [S20]*
>
> *Governance retrofit*
>
> *An SVP-driven 90-day safety reset across ~335 systems; mandatory two-person sign-off and senior-engineer approval for AI-assisted production changes. [S20]*
>
> *The fix was a deterministic gate on an irreversible action — bounded autonomy, retrofitted after the fact. [inference]*
>
> *[S20] Coding Agent Horror Stories: The Agent That Deleted Production — Docker (2026). https://www.docker.com/blog/coding-agent-horror-stories-the-agent-that-deleted-production/*

Now the story that puts the whole thesis into one incident — a failure mode this room knows intimately, wearing an AI-agent costume. [inference] Amazon's Kiro agent hit a bug, weighed its options, and concluded the cleanest fix was to delete the production environment and rebuild it from scratch — AWS Cost Explorer, one region, down for roughly thirteen hours. [S20] Read the mechanism, not the headline, because the headline invites the wrong lesson. [inference] It did not fail because it was a weak model — deleting and rebuilding is, abstractly, a valid repair. [inference] It failed because it had authority over an irreversible action and nothing deterministic stood between the decision and the effect — a permission-scoping failure, and everyone here has shipped the human version of it. [inference] Now watch the response, because the response is the thesis: Amazon did not go looking for a smarter agent; they ran an SVP-driven ninety-day safety reset across roughly three hundred and thirty-five of their most important systems and retrofitted mandatory two-person sign-off and senior-engineer approval for AI-assisted production changes. [S20] Bounded autonomy, added after the outage instead of before it. [inference] The whole talk in one sentence: put the gate on the irreversible action first, not after the thirteen-hour incident teaches you to. [inference] Let that one sit.

# 42 42. It generalises beyond software

**On screen:**

> *Harness*
>
> *Different industry, same law. [S33]*
>
> - *Salesforce runs "guided determinism": an agent graph where "some transitions are guarded by hard validation gates that have to pass before execution can continue." [S33]*
> - *The probabilistic model reasons; deterministic gates decide what is allowed to happen. [S33]*
>
> *[S33] Building enterprise AI agents that are both autonomous and reliable — Salesforce Engineering.*

One more example, from a different industry, so you do not file this under "coding-agent problem" and set it aside. [inference] Salesforce's production enterprise runtime uses a pattern they call "guided determinism": a business process modelled as a graph, where "some transitions are guarded by hard validation gates that have to pass before execution can continue." [S33] But whether that step is allowed to happen is decided by a deterministic validation gate on the transition, not by the model's confidence that it should. [S33] When the same move shows up independently in agent security, in compliance checking and in enterprise workflow, that is a sign it is a real engineering principle, not one vendor's marketing. [inference]

# 43 43. Your Monday

**On screen:**

> *Part V*
>
> *The "so what" for people who ship software, not papers.*

Part five — no claim on the divider — and the part you came for, so I have kept real time for it. [speculative] Everything up to here was the argument; this is the "so what" for people who ship software rather than write papers. [speculative] I will give you where the constraint moves, how the job changes, a concrete recipe, an honest-limits slide, and three things you could do on Monday — ending on the adopt-defer-refuse call you were hand-picked to make. [speculative]

# 44 44. The bottleneck moves downstream

**On screen:**

> *Monday*
>
> *"The binding constraint migrates downstream as generation gets cheap." [S21]*
>
> *+76%*
>
> *more PRs to review once agents ramped, Spotify Engineering [S31]*
>
> *The payoff is not fewer humans; it is redirecting human effort to review and verification. [inference]*
>
> *[S21] Harness Engineering — vault report · [S31] Coding is no longer the constraint — Spotify Engineering.*

Start with where the work goes, because this reframes the business case. [inference] As generation gets cheap, "the binding constraint migrates downstream" — off producing the artifact and onto understanding, validating, integrating and owning it. [S21] Here it is in a real number: Spotify's own engineering organisation reported 76% more pull requests to review once agents ramped. [S31] If you sold agents to leadership as headcount reduction, this number is the correction, and better to have it before the reorg than after. [inference]

# 45 45. The trade to watch

**On screen:**

> *Monday*
>
> - *Cheap generation moves the constraint to review — it does not remove it. [S21]*
> - *More PRs mean more surface area; quality debt surfaces downstream if review cannot keep pace. [inference]*
> - *The answer is the whole talk: deterministic gates and independent verification, so review scales without a human reading every line. [speculative]*
>
> *[S21] Harness Engineering — vault research report.*

So the trade to watch as you adopt this, stated as a risk not a promise. [inference] Cheapening generation moves the binding constraint downstream to review and integration; it does not delete the constraint, it relocates it. [S21] And if review capacity does not keep pace with the new volume, the quality debt does not disappear — it surfaces downstream, later, more expensively, in production. [inference] The way out is not "review harder" or "hire an army of reviewers." [inference] It is the whole talk applied to your own pipeline: deterministic gates and independent verification, so review scales without a human reading every line the agents produced. [inference] Make the machine check what the machine can check, and spend your scarce human attention on the judgement calls a gate cannot make. [inference]

# 46 46. The job shifts

**On screen:**

> *Monday*
>
> > *"The engineer's value is defined not by what the agent generates but by the control plane the engineer builds to constrain, verify, and absorb it." [S23]*
>
> *That moves you up the stack, not out of it. [speculative]*
>
> *[S23] Harness Engineering — practitioner synthesis — vault research report.*

Which changes the job description — in your favour, I think, though I know that is the anxious question under all of this. [inference] The report's framing: "the engineer's value is defined not by what the agent generates but by the control plane the engineer builds to constrain, verify, and absorb it." [S23] It becomes the control plane: the gates, the checks, the verification, the architecture that takes probabilistic output and makes it safe to run. [S23] The senior engineers here have done the human version of that for years; this is the same skill, aimed at a new source of unreliable output. [inference]

# 47 47. A deterministic-first recipe

**On screen:**

> *Monday*
>
> - *"Make the deterministic component load-bearing, make the model the orchestrator around it, and bound the whole with calibrated human oversight." [S8]*
> - *Close every loop against an external oracle — a test, a solver, an independent gate — never the model that produced the work. [S14]*
> - *Deterministic first; model judge only where the target is genuinely subjective. [inference]*
>
> *[S8] 5 Pillars — practitioner synthesis · [S14] Loop Engineering — vault research reports.*

If you take one reusable recipe home, take this — the whole architecture in three clauses. [inference] "Make the deterministic component load-bearing, make the model the orchestrator around it, and bound the whole with calibrated human oversight." [S8] Deterministic part carries the weight; model drives; human owns the fence. [inference] And in every loop inside that architecture, close against an external oracle — a test, a solver, an independent gate — never against the model that produced the work. [S14] The priority ordering holds: deterministic checks first because they are reliable and cheap, a model judge only where the target is genuinely, irreducibly subjective. [inference] Notice the recipe says nothing about which model — on purpose, because that is the part that does not expire when the next model ships. [inference]

# 48 48. Advocate the idea, hedge the label

**On screen:**

> *Monday*
>
> - *The vocabulary is contested — "harness," "loop," "graph" engineering. [inference]*
> - *"Advocate the engineering idea; hedge the compound noun." [S26]*
> - *The durable idea — verifier-gated, externally-grounded, bounded autonomy — survives whatever the term becomes. [speculative]*
>
> *[S26] Loop Engineering — vault concept note.*

A short caveat on vocabulary, so you are not caught out when someone pushes on the words. [inference] The labels in this space — harness engineering, loop engineering, graph engineering — are genuinely unsettled, and each was contested within weeks of being coined. [inference] So the vault's guidance, which I pass on directly: "advocate the engineering idea; hedge the compound noun." [S26] Do not go to war for the term — you will lose, because the term will change. [inference] Go to war for the durable idea underneath: verifier-gated loops, external grounding, bounded autonomy. [inference] That idea survives whatever the industry calls it next quarter, and it is what you are actually adopting — sell the mechanism, not the buzzword. [inference]

# 49 49. What this evidence does not settle

**On screen:**

> *Honest limits*
>
> •
>
> *Model-authored policies*
>
> *[S18]*
>
> *LLM-written policies leave residual attack. [S18]*
>
> •
>
> *Harness can backfire*
>
> *[S28]*
>
> *One tool-format swap cost ~38 points. [S28]*
>
> •
>
> *Widening surface*
>
> *[S36]*
>
> *GhostJacking extends agentjacking (Aug 2026). [S36]*
>
> •
>
> *Verifier scope*
>
> *[S37]*
>
> *Certifies only what it can see. [S37]*
>
> •
>
> *No-oracle zone*

> *[S39]*
>
> *No complete oracle; partial checks only. [S38][S39]*

The most credible slide in a technical talk is the one about what the evidence does not settle, so here are five honest limits — I would rather raise them than have you raise them in Q&A. [inference] One: deterministic control is not a free zero once the model writes the policy — the residual attack comes back. [S18] Two: not every harness change helps, and one wrong tool-format choice cost thirty-eight points, so measure changes against an eval set. [S28] Three: the attack surface is widening, not narrowing — GhostJacking, disclosed this month, extends agentjacking by poisoning trusted logs and alerts. [S36] Four, and this one lands on my strongest number, so I raise it before you do: even that closed loop only fixes the errors its verifier can see — the study's own authors flag that a defect from structural topology rather than parameters left the parameter-level loop "powerless," so the guarantee is bounded by the scope of the oracle you build, not by the domain it came from. [S37] And five, the honest one a hostile expert will press: most of your production targets — is this refactor correct, is this summary good — have no cheap external oracle at all, so where none exists the honest architecture is bounded autonomy plus human review, and sometimes the right call is to refuse. [inference] Knowing which zone you are standing in — oracle-rich, oracle-scoped, or oracle-free — is itself part of the job. [inference]

# 50 50. Three moves

**On screen:**

> *Monday*
>
> *1*
>
> *Add a check*
>
> *Find one self-graded loop; give it an independent check. [S14]*
>
> *2*
>
> *Gate one action*
>
> *Put one deterministic gate on an irreversible action — a merge, a deploy, a write. [S8]*
>
> *3*
>
> *Check one figure*
>
> *Take one number you cite about AI; confirm it is corroborated, not single-source. [S30]*
>
> *None of these needs a better model; all of them are engineering. [inference]*

Three concrete moves you could start on Monday morning — small enough to do, pointed enough to matter. [inference] One: find one loop where the model grades its own work, and give it an independent check — a test, a schema, an isolated-context checker — just one, as proof to yourselves that it changes the failure rate. [S14] Two: put one deterministic gate in front of one irreversible action — a merge, a deploy, a write to production — the Kiro fix, applied before your Kiro incident. [S8] Three: take one number you personally repeat about AI, and check whether it is corroborated or single-source — the thing I did with the 85% figure. [S30] Which is the point: the reliability was never going to come from the model. [inference]

# 51 51. Takeaways — what follows from the thesis

**On screen:**

> *In closing*
>
> *1*
>
> *Relocate the truth*
>
> *Reliability is bought by relocating truth-determination out of the model. [S6]*
>
> *2*
>
> *Independence wins*
>
> *Independence — not iteration — is the load-bearing variable. [S10]*
>
> *3*
>
> *Determinism first*
>
> *Push determinism to the load-bearing control; bound the model inside it. [S16]*
>
> *4*
>
> *Autonomy last*
>
> *Unbounded autonomy is unbounded risk [S7], so autonomy is what you add last. [inference]*

The same four takeaways I opened with, now earned rather than asserted — this is the summary, and notice it is what follows from the thesis, not a recap of section titles. [inference] If reliability is bought by relocating truth-determination out of the model, then it follows that independence of the verifier — not how hard you iterate — is the load-bearing variable. [S6] [S10] It follows that you push determinism to the load-bearing control and bound the model inside it, at every layer of the harness. [S16] And it follows that because unbounded autonomy is unbounded risk [S7], autonomy is the property you add last, after the check is in place. [inference] Four consequences, one thesis. [inference] If you disagree with the conclusions, the honest place to attack is the thesis they descend from — and that is exactly the argument I want to have next. [inference]

# 52 52. The one line to keep

**On screen:**

> *One line*
>
> > *"The model is a commodity; the harness is the product." [S22]*
>
> *Reliability is engineered, not prompted. [speculative]*
>
> *[S22] Harness Engineering — practitioner synthesis — vault research report.*

If you forget everything else, keep this one line: "the model is a commodity; the harness is the product." [S22] The model is the commodity — it changes every few months, everyone has the same ones, and it is not where your durable advantage lives. [inference] The harness is the product — the control plane, the gates, the verification, the architecture — the engineering asset that makes a stochastic model behave well enough to trust in production. [S22] Which brings us back to where we started: reliability is engineered, not prompted. [inference] That is the claim I opened with; I have given you the evidence for it and the honest limits against it, and now I would genuinely like you to tell me where it breaks — because you are exactly the room that would know. [inference] Thank you. [inference]

# 53 53. References

**On screen:**


> *[S1] Building AI Systems: 5 Pillars — comprehensive report — vault research report.*
>
> *[S2] 5 Pillars — context engineering — vault research report.*
>
> *[S3] 5 Pillars — context rot — vault research report.*
>
> *[S4] 5 Pillars — grounding, three layers — vault research report.*
>
> *[S5] 5 Pillars — closed-loop external-verifier compliance result (56.8→98.6) — vault research report.*

*[S6] 5 Pillars — relocating truth-determination — vault research report.*

*[S7] 5 Pillars — bounded autonomy — vault research report.*

*[S8] 5 Pillars — practitioner synthesis — vault research report.*

*[S9] Loop Engineering — two loop archetypes — vault research report.*

*[S10] Loop Engineering — verifier independence — vault research report.*

*[S11] Loop Engineering — "Gödel with a leaderboard," narrowed to "the price is never zero" — vault research report.*

*[S12] Loop Engineering — self-attribution bias (scoped: 1 preprint, "in one setting") — vault research report.*

*[S13] Loop Engineering — rule-based more reliable, LLM more variable — vault research report.*

*[S14] Loop Engineering — practitioner synthesis — vault research report.*

*[S15] Loop Engineering — guarantee in the verifier — vault research report.*

*[S16] Harness Engineering — the one law — vault research report.*

*[S17] Harness Engineering — Progent, hand-written policy — vault research report.*

*[S18] Harness Engineering — Progent, autonomous-policy residual — vault research report.*

*[S19] Harness Engineering — agentjacking, execution-layer controls — vault research report.*

*[S20] Coding Agent Horror Stories: The Agent That Deleted Production (Amazon Kiro) — Docker (2026). https://www.docker.com/blog/coding-agent-horror-stories-the-agent-that-deleted-production/*

*[S21] Harness Engineering — binding constraint migrates downstream — vault research report.*

*[S22] Harness Engineering — model is a commodity, harness is the product — vault research report.*

*[S23] Harness Engineering — the control plane the engineer builds — vault research report.*

*[S24] Harness Engineering — vault concept note.*

*[S25] Harness Engineering — OpenAI ~1M lines / LangChain +13.7 — vault concept note.*

*[S26] Loop Engineering — advocate the idea, hedge the noun — vault concept note.*

*[S27] Intuit Scrapped Its AI Agent Architecture Twice in Four Months — VentureBeat, VB Transform 2026. https://venturebeat.com/orchestration/intuit-scrapped-its-own-ai-agent-architecture-twice-in-four-months-at-vb-transform-2026-its-ai-vp-called-that-the-fast-path*

*[S28] Is Grep All You Need? How Agent Harnesses Reshape Agentic Search — vault reading note.*

*[S29] Over 100 AI Coding Agents Taken Over Via Agentjacking — vault watch note.*

*[S30] 85% Agentjacking Success Rate Not Independently Replicated — vault watch note.*

*[S31] Coding Is No Longer the Constraint — Spotify Engineering. https://engineering.atspotify.com/2026/6/code-with-claude-coding-is-no-longer-the-constraint*

*[S32] Progent: Securing AI Agents with Privilege Control — arXiv 2504.11703. https://arxiv.org/abs/2504.11703*

*[S33] Building Enterprise AI Agents That Are Both Autonomous and Reliable — Salesforce Engineering. https://engineering.salesforce.com/building-enterprise-ai-agents-that-are-both-autonomous-and-reliable/*

*[S34] Harness Engineering: Leveraging Codex in an Agent-First World — OpenAI. https://openai.com/index/harness-engineering/*

*[S35] Improving Deep Agents with Harness Engineering — LangChain. https://blog.langchain.com/improving-deep-agents-with-harness-engineering/*

*[S36] ThreatsDay: GhostJacking AI Attacks — The Hacker News (Aug 2026). https://thehackernews.com/2026/08/threatsday-ghostjacking-ai-attacks.html*

*[S37] 5 Pillars — Grounding, the closed-loop boundary condition (verifier scope bounds the guarantee) — vault research report.*

*[S38] Property-Based Testing for LLMs — arXiv 2307.04346, via vault note Alternative Methods to LLM Evaluations. https://arxiv.org/abs/2307.04346*

*[S39] Metamorphic Prompt Testing for LLMs — arXiv 2406.06864, via vault note Alternative Methods to LLM Evaluations. https://arxiv.org/html/2406.06864v1*

*[S40] Canary & Shadow Deployment for LLM Apps — vault note Alternative Methods to LLM Evaluations.*

*[S41] Structural Grounding — vault concept note (three-layer grounding taxonomy).*

*[S42] Idempotency in Distributed Systems — vault concept note.*

*[S43] Bounded Autonomy in AI — vault concept note.*

*[S44] Harness Engineering — deletion cadence — vault concept note.*

*[S45] Physics-in-the-Loop: Validated CAD Engineering Design — Berger et al., IJCAI-ECAI 2026 (arXiv 2605.19717), via 5 Pillars report. https://arxiv.org/abs/2605.19717*

*[S46] Loop Engineering — the theoretical case (Gödel / Tarski) — vault research report.*

*[S47] Loop Engineering — two loop archetypes (self-audit study) — vault research report.*

Reference card — pointers only, no new claim. [inference] Every figure on these slides traces to a numbered entry in sources.md, and the vault reports carry the deeper source chains. [inference] The handout goes a layer deeper on each. [inference]